



**Generation Interconnection
System Impact Study Report
for
Queue Project AF1-040
GRATZ TAP 69 KV
1 MW Capacity / 20 MW Energy**

August 2020

Table of Contents

1	Introduction.....	3
2	Preface.....	3
3	General	3
4	Point of Interconnection.....	4
5	Cost Summary	4
6	Transmission Owner Scope of Work	5
7	Attachment Facilities Cost Estimate	5
8	Direct Connection Cost Estimate.....	6
9	Non-Direct Connection Cost Estimate.....	6
10	Schedule.....	6
11	Interconnection Customer Requirements.....	7
	11.1 PPL EU Interconnection Requirements.....	7
	11.2 IC Direct Transfer Trip (DTT) Requirements	7
12	Revenue Metering and SCADA Requirements	7
	12.1 PJM Requirements	7
	12.2 Interconnected Transmission Owner Requirements.....	7
13	Summer Peak Analysis	7
	13.1 Generation Deliverability	8
	13.2 Multiple Facility Contingency	8
	13.3 Contribution to Previously Identified Overloads.....	8
	13.4 Steady-State Voltage Requirements	8
	13.5 Potential Congestion due to Local Energy Deliverability.....	8
14	Light Load Analysis	8
15	Short Circuit Analysis.....	8
16	Stability and Reactive Power	9
17	Affected Systems	9
18	Attachment 1: One Line Diagram	10
19	Attachment 2: Project Location	11

1 Introduction

This System Impact Study has been prepared in accordance with the PJM Open Access Transmission Tariff, 205, as well as the System Impact Study Agreement between the Interconnection Customer (IC), and PJM Interconnection, LLC (PJM), Transmission Provider (TP). The Interconnected Transmission Owner (ITO) is PPL.

2 Preface

The intent of the System Impact Study is to determine a plan, with approximate cost and construction time estimates, to connect the subject generation interconnection project to the PJM network at a location specified by the Interconnection Customer. As a requirement for interconnection, the Interconnection Customer may be responsible for the cost of constructing: Network Upgrades, which are facility additions, or upgrades to existing facilities, that are needed to maintain the reliability of the PJM system. All facilities required for interconnection of a generation interconnection project must be designed to meet the technical specifications (on PJM web site) for the appropriate transmission owner.

In some instances an Interconnection Customer may not be responsible for 100% of the identified network upgrade cost because other transmission network uses, e.g. another generation interconnection or merchant transmission upgrade, may also contribute to the need for the same network reinforcement. The possibility of sharing the reinforcement costs with other projects may be identified in the Feasibility Study, but the actual allocation will be deferred until the System Impact Study is performed.

The System Impact Study estimates do not include the feasibility, cost, or time required to obtain property rights and permits for construction of the required facilities. The project developer is responsible for the right of way, real estate, and construction permit issues. For properties currently owned by Transmission Owners, the costs may be included in the study.

3 General

The Interconnection Customer (IC), **Momentum Energy Storage Partners, LLC**, has proposed a Storage generating facility located in Dauphin County, Pennsylvania. The installed facilities will have a total capability of 20 MW with 1 MW of this output being recognized by PJM as Capacity. The proposed in-service date for this project is December 01, 2020. This study does not imply a TO commitment to this in-service date.

Queue Number	AF1-040
Project Name	Gratz Tap 69 KV
State	Pennsylvania
County	Dauphin
Transmission Owner	PPL
MFO	20
MWE	20
MWC	1
Fuel	Storage
Basecase Study Year	2023

New Service Customers proposing queue projects that can feasibly be commercially operable prior to June 1st of the basecase study year are required to request interim deliverability analysis.

4 Point of Interconnection

AF1-040 will interconnect with the PPL transmission system via a new transmission line tap off the Gratz Tap 69 kV, 4.15 miles from the Gratz Tap bus and 0.52 miles from the Gratz bus.

The Point of Interconnection (POI) will be where the PPL EU tap line terminates (with insulators) at the first dead-end structure inside the IC substation.

5 Cost Summary

The AF1-040 project will be responsible for the following costs:

Description	Total Cost
Total Physical Interconnection Costs	\$ 1,317,000
Allocation towards System Network Upgrade Costs*	\$ 0
Total Costs	\$ 1,317,000

This cost excludes a Federal Income Tax Gross Up charges. This tax may or may not be charged based on whether this project meets the eligibility requirements of IRS Notice 88-129. If at a future date it is determined that the Federal Income Tax Gross charge is required, the Transmission Owner shall be reimbursed by the Interconnection Customer for such taxes.

*As your project progresses through the study process and other projects modify their request or withdraw, then your cost allocation could change.

Note 1: PJM Open Access Transmission Tariff (OATT) section 217.3A outline cost allocation rules. The rules are further clarified in PJM Manual 14A Attachment B. The allocation of costs for a network upgrade will start with the first Queue project to cause the need for the upgrade. Later queue projects will receive cost allocation contingent on their contribution to the violation and are allocated to the queues that have not closed less than 5 years following the execution of the first Interconnection Service Agreement which identifies the need for this upgrade.

6 Transmission Owner Scope of Work

The total physical interconnection costs is given in the table below:

Description	Total Cost
Attachment Facilities	\$ 971,100
Direct Connection Network Upgrade	\$ 0
Non Direct Connection Network Upgrades	\$ 345,900
Total Physical Interconnection Costs	\$ 1,317,000

PPL EU can accommodate this interconnection by constructing a transmission line tap with a motor operated switch off the Gratz 69 kV Tap. The PPL EU scope of work also includes remote end relay work at the Sunbury Substation.

Study Assumptions

- No major environmental, real estate, siting, or permitting issues
- IC is responsible for acquisition of easements, permits, and right of way for any Direct Connection Network Upgrades and Attachment Facilities per PPL EU standards and requirements.
- PPL EU will perform all grading, site preparation, and establish access roads for the PPL EU owned Attachment Facilities per PPL EU standards and requirements.

7 Attachment Facilities Cost Estimate

69 kV Transmission Line Tap

The Attachment Facilities will connect to the Gratz 69 kV Tap off the Dauphin – Pine Grove 69 kV line. This scope of work is based on the IC collector substation GPS Coordinates: 40.6148720, -76.7241950.

PPL EU work will consist of installing the following:

- Install 69 kV tap off the existing Gratz Tap 69 kV line to AF1-040 substation dead-end structure.
- Install Motor Operated Load Break Air Break switch (MOLBAB) on the tap.

The total preliminary cost estimate for the Attachment work is given in the table below. These costs do not include CIAC Tax Gross-up.

Description	Total Cost
69 kV Transmission Line Tap	\$ 971,100
Total Attachment Facility Costs	\$ 971,000

8 Direct Connection Cost Estimate

None.

9 Non-Direct Connection Cost Estimate

Gratz Tap 69 kV Modifications to tie in the AF1-040 Attachment Facilities

- Tie the new AF1-040 Attachment Facilities into the Gratz Tap 69 kV line
- Remove steel pole at 28488S46852 and install two pole DCT style tap.

Remote End Work – Sunbury Substation

- Complete remote end relay work at Sunbury 69 kV Substation for Direct Transfer Trip.
- Model IC in CAPE and conduct a wide area short-circuit study two busses away from the IC facilities. Identify affected relays and revise settings as needed.
- Conduct a review of the IC relay settings and engineering package (submitted by IC to PPL EU).

The total preliminary cost estimate for the Non-Direct Connection work is given in the table below. These costs do not include CIAC Tax Gross-up.

Description	Total Cost
Gratz Tap 69 kV Modifications to tie in the AF1-040 Attachment Facilities	\$ 107,900
Remote End Work – Sunbury Substation	\$ 238,000
Total Non-Direct Connection Facility Costs	\$ 345,900

10 Schedule

The estimated time to complete the scope of work is 12-18 months after the PJM three-party Interconnection Service Agreement (ISA) and Interconnection Construction Service Agreement (ICSA) are signed and PPL EU receives Notice to Proceed from the IC.

11 Interconnection Customer Requirements

11.1 PPL EU Interconnection Requirements

PPL EU applicable technical standards that address requirements for interconnection of generation, transmission, and end user facilities can be found at the following link:

<https://pjm.com/planning/design-engineering/to-tech-standards/private-ppl.aspx>

11.2 IC Direct Transfer Trip (DTT) Requirements

PPL EU will require an independent communication path, for Direct Transfer Trip (DTT) of the IC Intertie Protective Relaying (IPR) Fault Interrupting Devices (FIDs), consisting of one communication circuit with the Sunbury Substation.

PPL EU does not have OPGW available on the Gratz Tap 69 kV line available for DTT to the Sunbury Substation. PPL EU assumes that the IC will procure the independent communication path through a third-party provider. Upon request, PPL EU will evaluate the feasibility of installing OPGW on the Gratz Tap 69 kV line for DTT.

12 Revenue Metering and SCADA Requirements

12.1 PJM Requirements

The Interconnection Customer will be required to install equipment necessary to provide Revenue Metering (KWH, KVARH) and real time data (KW, KVAR) for IC's generating Resource. See PJM Manuals M-01 and M-14D, and PJM Tariff Section 8 of Attachment O.

12.2 Interconnected Transmission Owner Requirements

Installation of revenue grade Bi-directional Metering Equipment will be required in the vicinity of the POI to measure kWh and kVARh. PPL EU will design and supply the required metering equipment; all installation costs would be borne by the IC including CTs/PTs. All metering equipment must meet applicable PPL EU tariff requirements as well as being compliant with all applicable requirements of the PJM agreements. The equipment must provide bidirectional revenue metering (kWh and kVARh) and real-time data (kW, kVAR, circuit breaker status, and generator bus voltages) for the IC's generating resource. The metering equipment should be housed in a control cabinet or similar enclosure and must be accessible to PPL EU metering personnel.

13 Summer Peak Analysis

The Queue Project AF1-040 was evaluated as a 20 MW (Capacity 1.0 MW) injection at the Gratz 69 kV substation in the PPL area. Project AF1-040 was evaluated for compliance with applicable reliability planning criteria (PJM, NERC, NERC Regional Reliability Councils, and Transmission Owners). Project AF1-040 was studied with a commercial probability of 100.0 %. Potential network impacts were as follows:

13.1 Generation Deliverability

(Single or N-1 contingencies for the Capacity portion only of the interconnection)

None

13.2 Multiple Facility Contingency

(Double Circuit Tower Line, Fault with a Stuck Breaker, and Bus Fault contingencies for the full energy output)

None

13.3 Contribution to Previously Identified Overloads

(This project contributes to the following contingency overloads, i.e. "Network Impacts", identified for earlier generation or transmission interconnection projects in the PJM Queue)

None

13.4 Steady-State Voltage Requirements

None

13.5 Potential Congestion due to Local Energy Deliverability

PJM also studied the delivery of the energy portion of this interconnection request. Any problems identified below are likely to result in operational restrictions to the project under study. The developer can proceed with network upgrades to eliminate the operational restriction at their discretion by submitting a Merchant Transmission Interconnection request.

Note: Only the most severely overloaded conditions are listed below. There is no guarantee of full delivery of energy for this project by fixing only the conditions listed in this section. With a Transmission Interconnection Request, a subsequent analysis will be performed which shall study all overload conditions associated with the overloaded element(s) identified.

None

14 Light Load Analysis

Light Load Analysis is not required for this project.

15 Short Circuit Analysis

The following Breakers are overdutied:

None

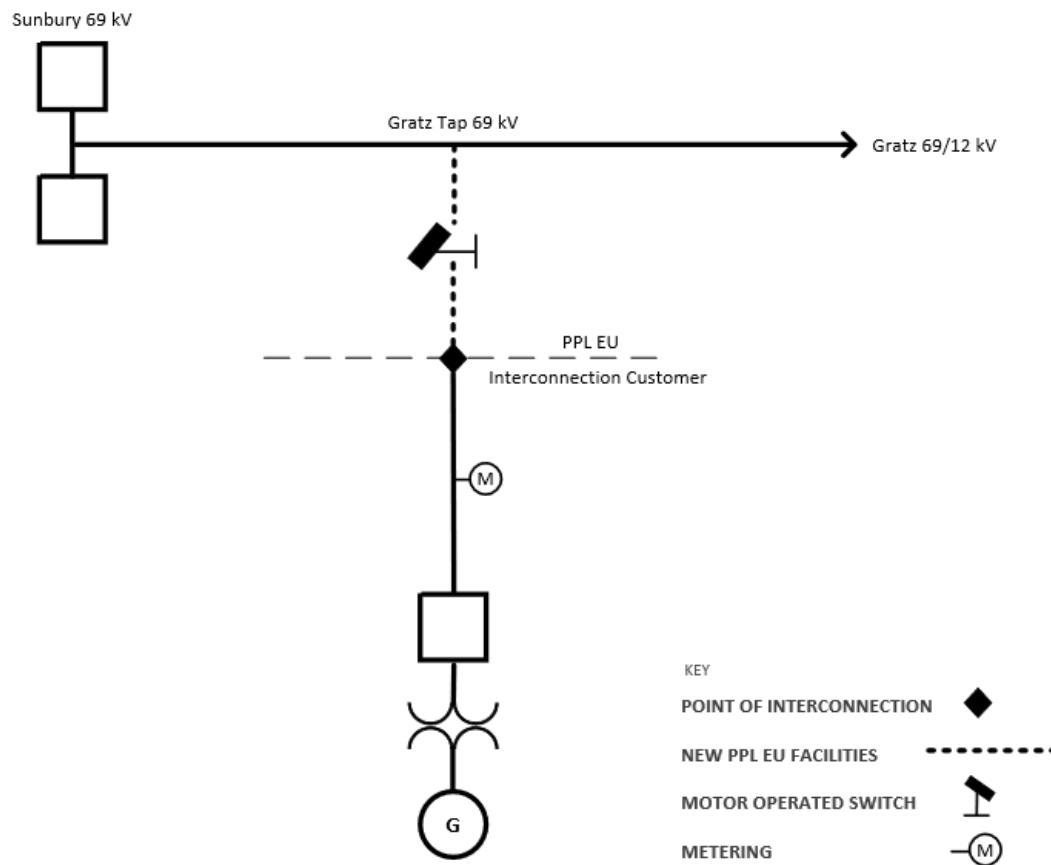
16 Stability and Reactive Power

Stability Analysis will be performed during the Facilities Study. Additional system reinforcements may be identified to mitigate any stability concerns.

17 Affected Systems

None

18 Attachment 1: One Line Diagram



19 Attachment 2: Project Location

